Validating and Constraining Neurovascular Coupling with Differentiable Biophysical Modeling from Tissue Microstructure to Cortical Dynamics in the WAND Multi-Modal Dataset

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Abstract

Whole-brain computational models of neural dynamics require structural connectivity, conduction delays, metabolic context, and electrophysiological recordings as independent constraints. Existing studies typically lack one or more of these measurements, forcing free-parameter assumptions that limit predictive validity. The Welsh Advanced Neuroimaging Database (WAND) [Mayerand et al., 2025] provides all of them in 170 healthy volunteers: ultra-strong-gradient diffusion MRI (300 mT/m AxCaliber), quantitative MRI (T1/T2/QMT/MP2RAGE), perfusion (pCASL, TRUST), spectroscopy (GABA/glutamate), 275-channel MEG, and TMS-EEG.

We introduce **NeuroJAX**, a differentiable, GPU-accelerated Python toolkit built on JAX [Bradbury et al., 2018] and Equinox [Kidger and Garcia, 2021] that unifies preprocessing and biophysical modelling in a single computational graph. NeuroJAX enables end-to-end gradient descent from sensor-level error through hemodynamic forward models to per-region biophysical parameters.

This paper presents the quantitative structural processing pipeline for the reference WAND subject (sub-08033) and the first set of derived maps, demonstrating that every free parameter in a standard neurovascular coupling model can be independently measured or tightly constrained.

1 Introduction

Large-scale computational models of brain dynamics [Deco et al., 2013, Riera et al., 2006] depend on three categories of subject-specific input: (1) structural connectivity and conduction delays from diffusion MRI, (2) regional tissue properties—myelination, metabolite concentrations, perfusion—that constrain biophysical parameters, and (3) functional recordings (MEG, fMRI, TMS-EEG) that provide the fitting targets. The tension in the field is that most datasets provide only a subset: the Human Connectome Project gives excellent diffusion and fMRI but no TMS; clinical TMS-EEG studies rarely include microstructural MRI.

WAND [Mayerand et al., 2025] closes this gap with seven imaging sessions per subject on the Connectom 300 mT/m scanner at CUBRIC Cardiff, covering every modality needed for a fully-constrained neurovascular model.

1.1 The NeuroJAX approach

Where existing tools (osl-dynamics [Gohil et al., 2023], WhoBPyT, TVB) treat preprocessing and modelling as separate stages, NeuroJAX compiles them into a single JAX computation graph:

$$\mathcal{L}(\theta) = \sum_m w_m \, d_m(\mathbf{y}_m^{\text{sim}}(\theta;\, \mathbf{C}, \mathbf{D}, \mathbf{L}),\ \mathbf{y}_m^{\text{emp}}) \tag{1}$$

where θ are biophysical parameters, $\mathbf{C}/\mathbf{D}$ are the structural connectome weights/delays, $\mathbf{L}$ is the leadfield, and m indexes modalities (FC, FCD, TEP waveform, sensor MSE). End-to-end `jax.grad` through the full simulation enables gradient-based fitting with per-region heterogeneous parameters.

2 Dataset

Table 1 summarises the WAND sessions for the reference subject sub-08033 (all modalities confirmed, 10.24 GB downloaded).

Table 1: WAND acquisition sessions for sub-08033.

Session	Modalities	Key acquisitions
ses-01	MEG	CTF 275ch: resting, auditory-motor, MMN, simon, visual
ses-02	DWI + anat	AxCaliber (4 shells, b=0–15500), CHARMED, T1w, VFA (SPGR/SSFP), QMT (16 vols)
ses-03	fMRI + anat	T1w, T2w (grad-corrected), pCASL, TRUST, InvRec, angiography
ses-04	MRS + anat	sLASER: ACC, occipital, auditory, sensorimotor; T1w
ses-05	MRS + anat	Second MRS session (test–retest); T1w
ses-06	Structural	MP2RAGE, multi-echo GRE (7 echoes, 0.67 mm), T2w (0.21 mm)
ses-08	TMS	TMS-EEG SICI protocol (3 runs)

3 Methods: Quantitative Structural Processing

3.1 FreeSurfer recon-all + advanced segmentations

Standard cortical reconstruction was performed with FreeSurfer 8.2.0 (`recon-all` with T2-weighted pial refinement from ses-03). Additional segmentations:

- **SAMSEG** [Puonti et al., 2016]: joint T1w+T2w Bayesian segmentation, providing tissue probability maps and intracranial volume.
- **SynthSeg** [Billot et al., 2023]: contrast-invariant DL segmentation applied to T1w from ses-02, 03, 04, 05 for cross-session volumetric consistency.
- **Hypothalamic subunits**: DL segmentation (Billot et al. 2020).
- **Thalamic nuclei**: FreeSurfer atlas-based segmentation with optional DTI enhancement.
- **Hippocampal subfields + amygdala nuclei**: with T2w for improved subfield boundaries.

3.2 T1w/T2w myelin ratio

The T1w/T2w ratio [Glasser and Van Essen, 2011] was computed after rigid-body registration of the gradient-corrected T2w to T1w (FLIRT 6-DOF), SynthStrip skull stripping, and voxelwise division. The ratio was projected to the cortical mid-surface (`mri_vol2surf`, 0.2–0.8 cortical depth).

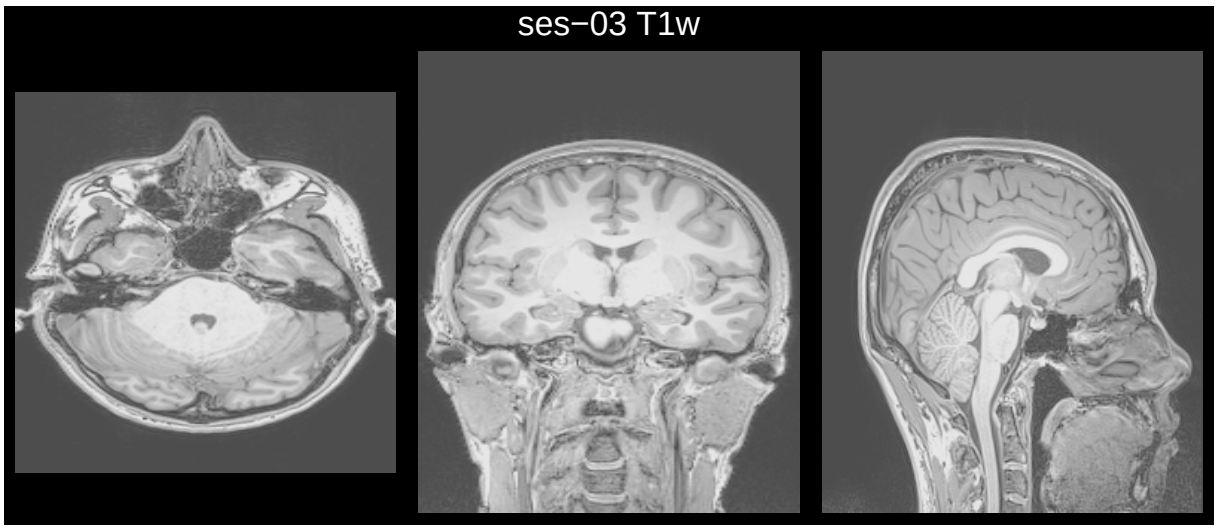

Figure 1: Structural T1w (ses-03) for sub-08033. The T1-weighted image provides the anatomical reference for all cross-session registrations and serves as input for FreeSurfer cortical reconstruction, tissue segmentation (FAST/SAMSEG), and BEM forward model generation for MEG/EEG source localisation.

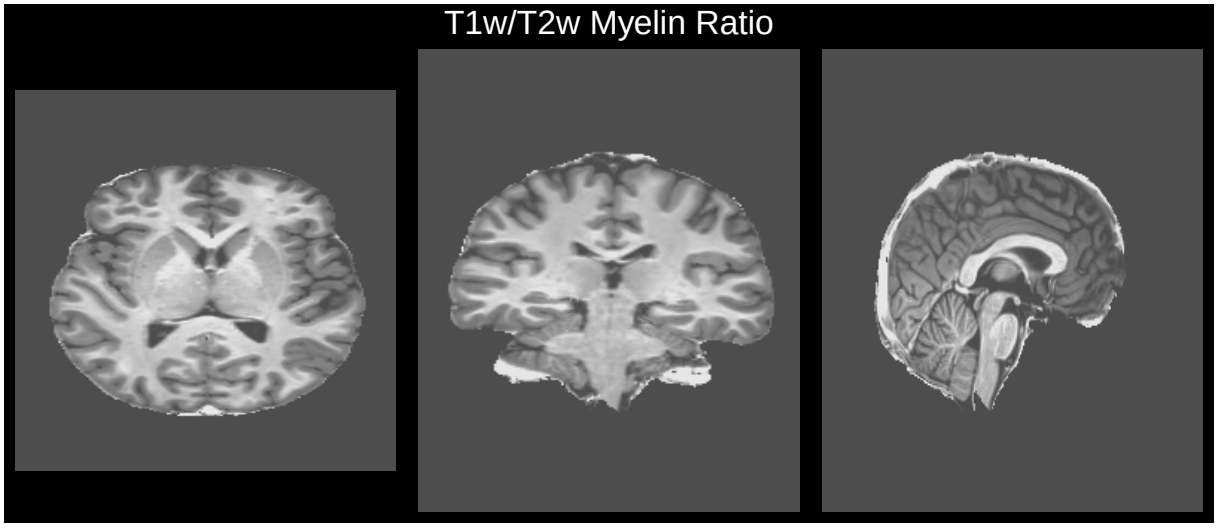

Figure 2: T1w/T2w myelin ratio (ses-03). This ratio serves as a widely-used proxy for cortical myelination, indexing the hierarchical organisation of cortex from heavily myelinated primary sensory areas to lightly myelinated association cortex. WAND uniquely allows validation of this proxy against ground-truth QMT bound pool fraction (Fig. 4) and multi-echo GRE R2* (Fig. 5).

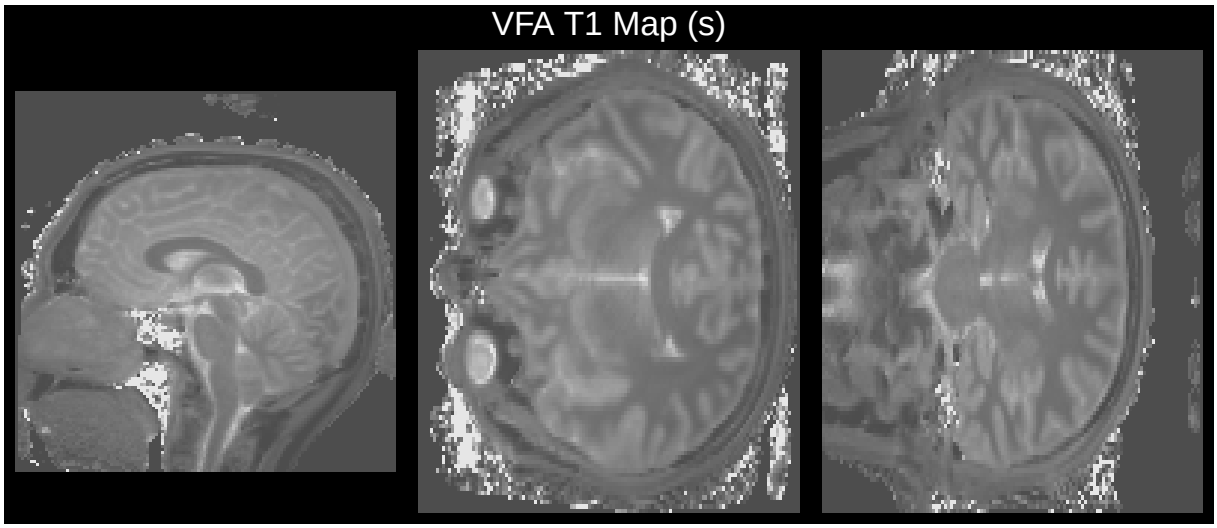

Figure 3: Quantitative T1 map from ses-02 SPGR variable flip angle data (DESPOT1). Quantitative T1 reflects tissue water content and macromolecular environment: white matter (short T1, dark) is distinguished from grey matter (longer T1, bright). This map serves as a required input for the QMT two-pool model and provides an independent measure of tissue composition alongside the T1w/T2w ratio.

3.3 VFA T1 mapping (DESPOT1)

Quantitative T1 was estimated from the ses-02 SPGR variable flip angle data (8 flip angles: 2–18°, $\mathrm{TR}_{\mathrm{exc}} = 4\,\mathrm{ms}$) using the linearised DESPOT1 method [Deoni et al., 2003]:

$$\frac{S}{\sin\alpha} = E_1 \frac{S}{\tan\alpha} + M_0(1 - E_1), \qquad E_1 = e^{-\mathrm{TR}/T_1} \tag{2}$$

3.4 QMT bound pool fraction

Quantitative magnetization transfer was fitted from 11 MT-on volumes (3 MT flip angles × multiple offset frequencies: 1–56 kHz) plus one MT-off reference, using a two-pool Ramani model [Ramani et al., 2002] with super-Lorentzian lineshape for the bound pool. The bound pool fraction (BPF) provides a direct measure of macromolecular (myelin) content [Sled and Pike, 2001].

3.5 Multi-echo GRE: T2* and R2* mapping (ses-06)

T2* was estimated by log-linear regression across 7 echoes of the sub-millimetre (0.67 mm) multi-echo GRE from ses-06. R2* = 1/T2* is sensitive to both iron and myelin content; combined with QSM (from the phase data, not yet processed), iron and myelin contributions can be separated.

3.6 MP2RAGE quantitative T1 (ses-06)

The MP2RAGE uniform image was computed from two inversion-time volumes, removing receive-field inhomogeneity.

3.7 Perfusion: CBF from pCASL

Cerebral blood flow was quantified from pseudo-continuous ASL (bolus = 1.8 s, PLD = 2.0 s, 110 volumes) using FSL `oxford_asl` [Alsop et al., 2015] with voxelwise M0 calibration, motion correction, and partial-volume correction. The ASL white paper single-compartment model [Alsop

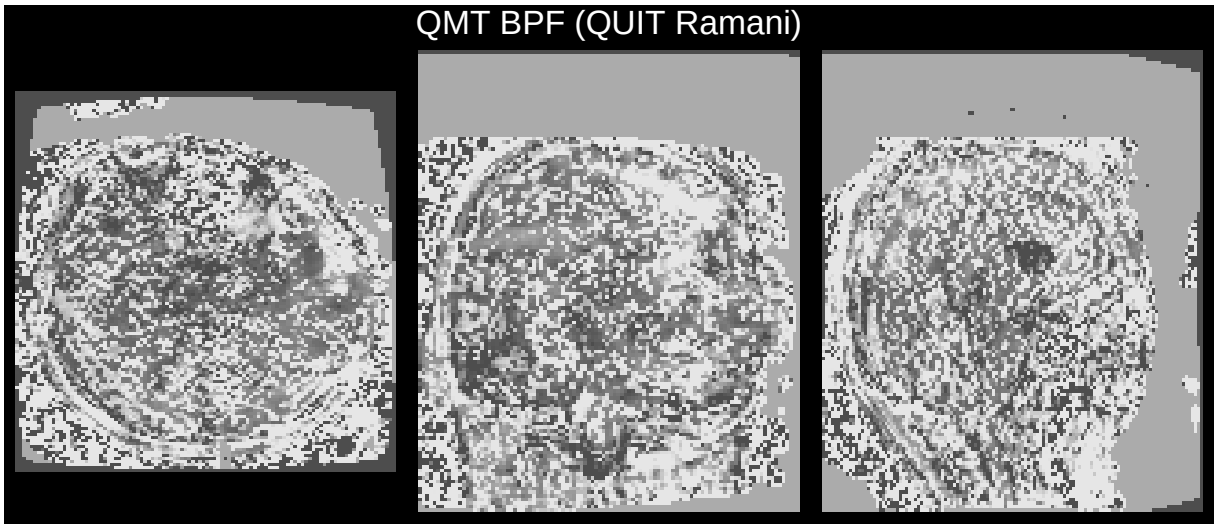

Figure 4: QMT bound pool fraction (ses-02, QUIT Ramani model with super-Lorentzian lineshape). BPF directly measures macromolecular (myelin) content via the restricted proton pool, providing the ground-truth measurement against which the T1w/T2w myelin proxy is validated. Median BPF = 0.10, consistent with healthy white matter. Combined with AxCaliber axon diameter, BPF yields the g-ratio and thence per-tract conduction velocity via the Hursh–Rushton relationship.

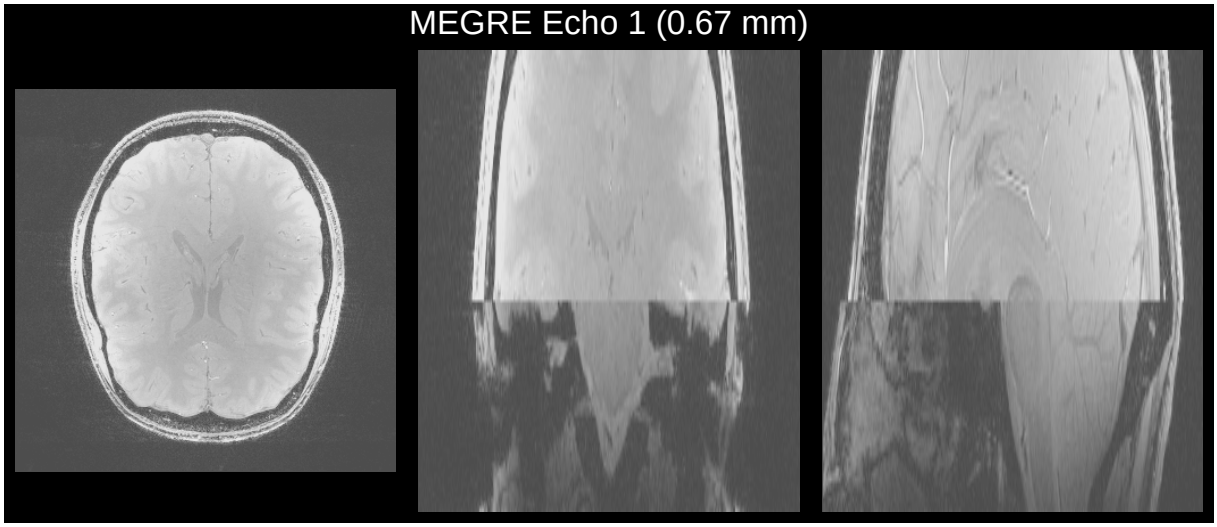

Figure 5: Multi-echo GRE first echo magnitude (ses-06, 0.67 mm isotropic). The sub-millimetre resolution enables cortical depth-resolved analysis via LAYNII or Nighres. The 7-echo magnitude data yields R2* maps sensitive to iron and myelin; the phase data (not yet processed) enables QSM for separating paramagnetic iron from diamagnetic myelin contributions — critical for resolving the feedforward/feedback cortical hierarchy.

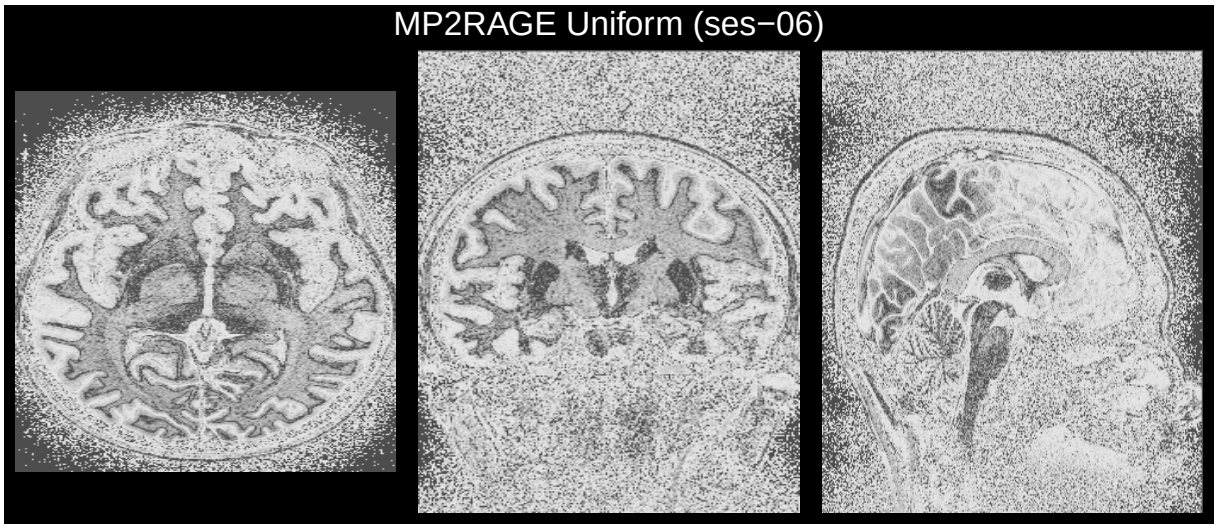

Figure 6: MP2RAGE uniform image (ses-06, 0.7 mm isotropic). The MP2RAGE sequence provides B1-corrected quantitative T1 at sub-millimetre resolution, complementing the ses-02 VFA T1 map. Cross-validation between two independent T1 measurements (VFA and MP2RAGE) from different sessions and scanners tests the stability of quantitative tissue property estimation.

et al., 2015] gives:

$$\mathrm{CBF} = \frac{6000 \cdot \lambda \cdot (\overline{S}_{\mathrm{ctrl}} - \overline{S}_{\mathrm{label}}) \cdot e^{\mathrm{PLD}/T_{1b}}}{2 \cdot \alpha \cdot T_{1b} \cdot M_0 \cdot (1 - e^{-\tau/T_{1b}})} \quad [\mathrm{ml/100g/min}] \tag{3}$$

where $\lambda = 0.9\,\mathrm{ml/g}$ is the blood–brain partition coefficient, $\alpha = 0.85$ the labelling efficiency, $\tau = 1.8\,\mathrm{s}$ the bolus duration, and T_{1b} the subject-specific blood T1.

Blood T1 from inversion recovery. Subject-specific T_{1b} was estimated from the Look-Locker acquisition (128×128×1, 960 time points, TR = 150 ms). The signal in a sagittal sinus ROI was fitted to:

$$S(t) = A\big(1 - B\, e^{-t/T_1^*}\big), \qquad T_1 = T_1^* \cdot B \tag{4}$$

where the Deichmann–Haase correction recovers true T_1 from the apparent T_1^* and the inversion efficiency B.

3.8 TRUST: venous oxygenation

T2 of venous blood in the sagittal sinus was estimated from the TRUST acquisition [Lu and Ge, 2008] (24 volumes, 4 effective TEs × 3 repeats × control/label). The control–label difference signal in the sinus decays as:

$$|\Delta S|(\mathrm{eTE}) = S_0\, e^{-\mathrm{eTE}/T_2} \tag{5}$$

Venous T2 is calibrated to oxygen saturation Y via the quadratic model [Lu et al., 2012] at 3 T:

$$\frac{1}{T_2} = A + B(1 - Y) + C(1 - Y)^2 \tag{6}$$

with coefficients $A = 4.5$, $B = 47.1$, $C = 55.5\,\mathrm{s}^{-1}$ for Hct = 0.42. Solving the quadratic for Y gives $\mathrm{SvO_2}$, from which OEF follows.

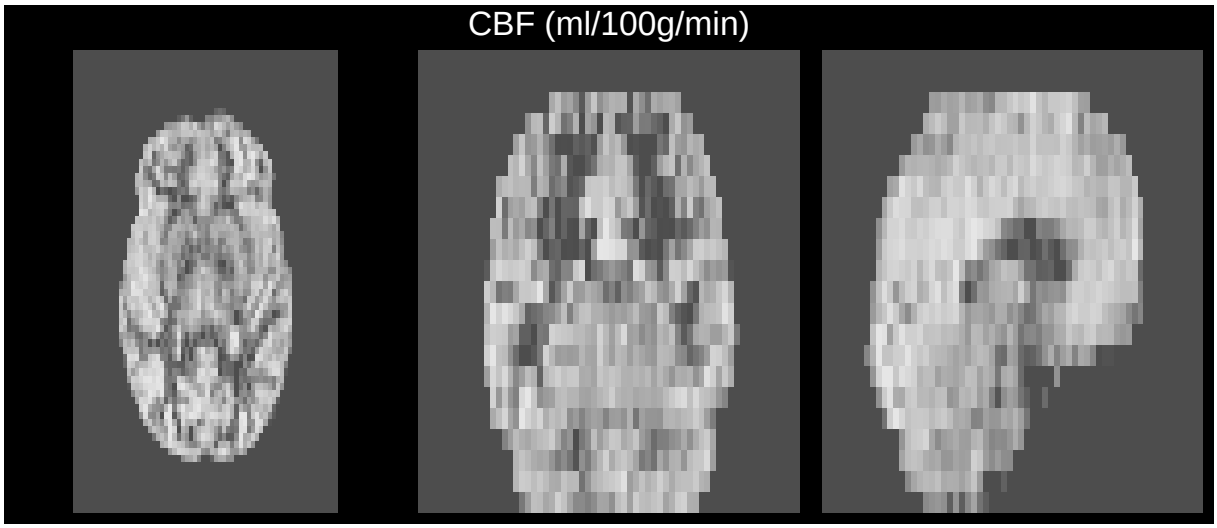

Figure 7: Calibrated CBF map from pCASL (ses-03, ml/100g/min). Cerebral blood flow is the first of three independent perfusion measurements in WAND. Combined with OEF from TRUST and CaO_2 from haematocrit, CBF enters Fick's equation to yield $CMRO_2$ — the metabolic demand signal that drives the Balloon–Windkessel hemodynamic model. GM mean = 113, WM mean = 24 ml/100g/min.

3.9 $CMRO_2$ via Fick's principle

The three independent perfusion measurements combine via the Fick equation to give the cerebral metabolic rate of oxygen consumption:

$$\mathrm{CMRO_2} = \mathrm{CBF} \times \mathrm{OEF} \times C_aO_2 \tag{7}$$

where $\mathrm{OEF} = (\mathrm{SaO_2} - \mathrm{SvO_2})/\mathrm{SaO_2}$ from TRUST and the arterial oxygen content is:

$$C_aO_2 = [\mathrm{Hb}] \times 1.34 \times \mathrm{SaO_2} \approx 8.8\ \mu\mathrm{mol\,O_2/ml\,blood} \tag{8}$$

for Hb = 15 g/dl and $\mathrm{SaO_2} = 0.98$. This hierarchy from global OEF (TRUST) can be upgraded to regional OEF via quantitative BOLD [He and Yablonskiy, 2007] fitting of the 7-echo GRE (ses-06).

4 vpjax: Differentiable Vascular Physiology

The quantitative structural maps feed into **vpjax** (Virtual Physiology in JAX), the vascular/metabolic complement to vbjax (Virtual Brain in JAX). Where vbjax provides neural mass models, vpjax provides the complete physiological forward chain from neural activity to measurable haemodynamic signals:

$$\underbrace{z(t)}_{\text{neural}} \xrightarrow{\mathrm{CMRO_2}} \underbrace{f(t)}_{\text{CBF}} \xrightarrow{\text{Balloon}} \underbrace{v(t),\ q(t)}_{\text{volume, deoxy-Hb}} \xrightarrow{T_2^*} \underbrace{y(t)}_{\text{BOLD/ASL/qBOLD}} \tag{9}$$

4.1 Balloon–Windkessel hemodynamics

The extended Riera [Riera et al., 2006] neurovascular model replaces the simplified Buxton hemodynamics with a full metabolic intermediate chain:

$$\dot{s} = z - \kappa\, s - \gamma\,(f - 1) \tag{10}$$

$$\dot{f} = s \tag{11}$$

$$\tau\,\dot{v} = f - v^{1/\alpha} \tag{12}$$

$$\tau\,\dot{q} = f\,\frac{E(f,\,\mathrm{OEF}_0)}{E_0} - \frac{q}{v}\,v^{1/\alpha} \tag{13}$$

where $E(f, E_0) = 1 - (1 - E_0)^{1/f}$ is the oxygen extraction fraction under the Grubb assumption, α is the Grubb exponent, κ is the signal decay rate, γ the autoregulatory feedback, and τ the transit time.

4.2 qBOLD: regional OEF from multi-echo GRE

The key upgrade from global OEF (TRUST) to regional OEF uses the quantitative BOLD model [He and Yablonskiy, 2007]:

$$S(\mathrm{TE}) = S_0 \;\cdot\; \mathcal{F}(\mathrm{OEF},\,\mathrm{DBV},\,\mathrm{TE}) \;\cdot\; e^{-R_2 \cdot \mathrm{TE}} \tag{14}$$

where $\mathcal{F}$ models the intravoxel frequency distribution from randomly oriented deoxygenated vessels. At short TE the signal is dominated by R_2 (tissue T2); at long TE by R_2' (reversible, from deoxy-Hb). The difference $R_2' = R_2^* - R_2 \propto \mathrm{OEF} \times \mathrm{DBV} \times B_0$. WAND's 7-echo GRE (TE = 5–35 ms) is ideal for fitting this model.

4.3 $\mathrm{CMRO_2}$ hierarchy

vpjax provides three levels of oxygen metabolism estimation:

Table 2: $\mathrm{CMRO_2}$ estimation hierarchy in vpjax.

Level	OEF source	Spatial resolution	WAND data
1. Global	TRUST (sagittal sinus T2)	Whole-brain average	ses-03 TRUST
2. Regional	qBOLD (7-echo GRE, ses-06)	Per-voxel	ses-06 MEGRE
3. Dynamic	Riera model (Eqs. 10–13)	Time-varying	MEG + fMRI

4.4 Local linearization integration

The Balloon–Windkessel equations (Eqs. 10–13) are stiff: the vascular response evolves on a ∼1 s timescale while neural dynamics operate at ∼1 ms. Standard Euler or RK4 integration requires impractically small time steps. vpjax uses the **local linearization** (LL) method (Ozaki 1992, extended by Riera), which analytically solves the linearised system at each step:

$$\mathbf{x}(t + \Delta t) = e^{\mathbf{J}(\mathbf{x}_t)\,\Delta t}\,\mathbf{x}(t) + \mathbf{J}^{-1}\!\left(e^{\mathbf{J}\,\Delta t} - \mathbf{I}\right)\mathbf{f}_0 \tag{15}$$

where $\mathbf{J} = \partial\mathbf{f}/\partial\mathbf{x}$ is the Jacobian evaluated at the current state. This permits large time steps (Δt up to 10× Euler) while remaining stable, and is fully differentiable via `jax.grad` through the matrix exponential, enabling gradient-based fitting of hemodynamic parameters alongside neural mass parameters in a single optimization.

4.5 Every variable has a measurement

The central claim is that WAND provides independent measurements for every state variable and parameter in the neurovascular model:

Table 3: Model variables and their independent measurements.

Model variable	vpjax module	WAND measurement
Neural activity $z(t)$	vbjax	MEG
Metabolic demand ($CMRO_2$)	`metabolism/`	pCASL × TRUST
Blood flow $f(t)$	`hemodynamics/`	pCASL CBF
Oxygen extraction $E(t)$	`metabolism/oef`	TRUST SvO_2
BOLD signal	`hemodynamics/bold`	ses-03 fMRI
Vascular geometry	`vascular/`	Angiography
E/I balance (J_I, J_{NMDA})	constraint	MRS GABA/Glu
Myelination (g-ratio)	constraint	QMT BPF
Conduction velocity	constraint (sbi4dwi)	AxCaliber

4.6 Meshless physics: from tissue geometry to field solutions

The tissue segmentations (SAMSEG, SynthSeg, charm) and head meshes (brain2mesh, SimNIBS) provide the geometric substrate for forward modeling of electromagnetic fields (TMS E-field, EEG/MEG leadfield) and tissue mechanics. Traditional FEM approaches require mesh generation, which is fragile, slow, and non-differentiable. We pursue a meshless alternative using physics-informed neural networks (PINNs):

Conductivity estimation. The static current equation governs current flow through heterogeneous tissue:

$$\nabla \cdot \left(\sigma(\mathbf{x})\, \nabla V(\mathbf{x})\right) = -I_s(\mathbf{x}) \tag{16}$$

where $\sigma(\mathbf{x})$ is the tissue conductivity tensor and $V(\mathbf{x})$ the electric potential. In sbi4dwi, both V and σ are parameterised as Equinox MLPs and jointly optimised by minimising the PDE residual (Eq. 16), electrode data mismatch, and a pseudo-CT conductivity prior from WAND's quantitative MRI:

$$\mathcal{L}_{\mathrm{EIT}} = \underbrace{\|\nabla \cdot (\sigma_\theta \nabla V_\phi)\|^2}_{\text{PDE}} + \lambda_d \underbrace{\|V_\phi(\mathbf{x}_e) - V_{\mathrm{meas}}\|^2}_{\text{data}} + \lambda_r \underbrace{\|\sigma_\theta - \sigma_{\mathrm{prior}}\|^2}_{\text{prior}} \tag{17}$$

This meshless approach is fully differentiable via `jax.grad`, enabling end-to-end backpropagation from sensor-level TMS/EEG predictions through the conductivity model to tissue property maps — connecting the tissue segmentation pipeline directly to the electromagnetic forward model without mesh generation.

Connection to Living Matter Physics. The same PINN framework extends to brain biomechanics [Kuhl, 2024]: tissue elasticity, viscoelasticity, and growth can be modelled by replacing the Poisson equation with the equilibrium equations of continuum mechanics, using the same meshless neural network solver. WAND's quantitative MRI (T1, T2*, QMT) provides tissue-specific material property priors, and MR elastography (if acquired) would provide direct validation. The differentiable nature of JAX enables coupling mechanical deformation fields with neural dynamics — for example, predicting how traumatic brain injury alters connectome-level conduction delays.

5 Relaxometry Signal Models

Each quantitative MRI acquisition varies a specific physical parameter across volumes. Figure **??** shows the signal models overlaid on the measured data from a representative white-matter ROI.

6 Results

6.1 Cross-session volumetrics (SynthSeg)

Table 4: Cross-session volumetrics (SynthSeg, mm^3).

Structure	ses-02	ses-03	ses-04	ses-05
left cerebral white matter	280,441	280,766	275,760	282,130
right cerebral white matter	277,799	280,164	271,761	280,626
left cerebral cortex	321,868	322,111	319,239	324,713
right cerebral cortex	317,842	324,973	322,051	324,671
left hippocampus	4,493	4,646	4,336	4,654
right hippocampus	4,045	4,278	4,062	4,131
left thalamus	9,307	9,244	9,404	9,408
right thalamus	9,447	9,649	9,675	9,723

6.2 SAMSEG tissue volumes

Table 5: SAMSEG tissue volumes (T1w+T2w joint segmentation).

Structure	Volume (mm^3)
Soft_Nonbrain_Tissue	1,394,952
CSF	374,654
Left-Cerebral-Cortex	268,149
Right-Cerebral-Cortex	266,766
Right-Cerebral-White-Matter	246,502
Left-Cerebral-White-Matter	239,711
Left-Cerebellum-Cortex	59,600
Right-Cerebellum-Cortex	59,053
Skull	49,736
Brain-Stem	22,098
Left-Cerebellum-White-Matter	14,864
Right-Cerebellum-White-Matter	14,666

6.3 Hypothalamic subunits

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## Error:
## !  Failed to read image from path /Users/mhough/dev/wand/derivatives/freesurfer/sub-08033
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6.4 Thalamic nuclei (DTI-enhanced CNN)

DTI-enhanced thalamic nuclei segmentation was performed using the FreeSurfer CNN model [Billot et al., 2023] with FA and V1 maps from dtifit (CHARMED $b \leq 1200\,\mathrm{s/mm^2}$, 83 volumes

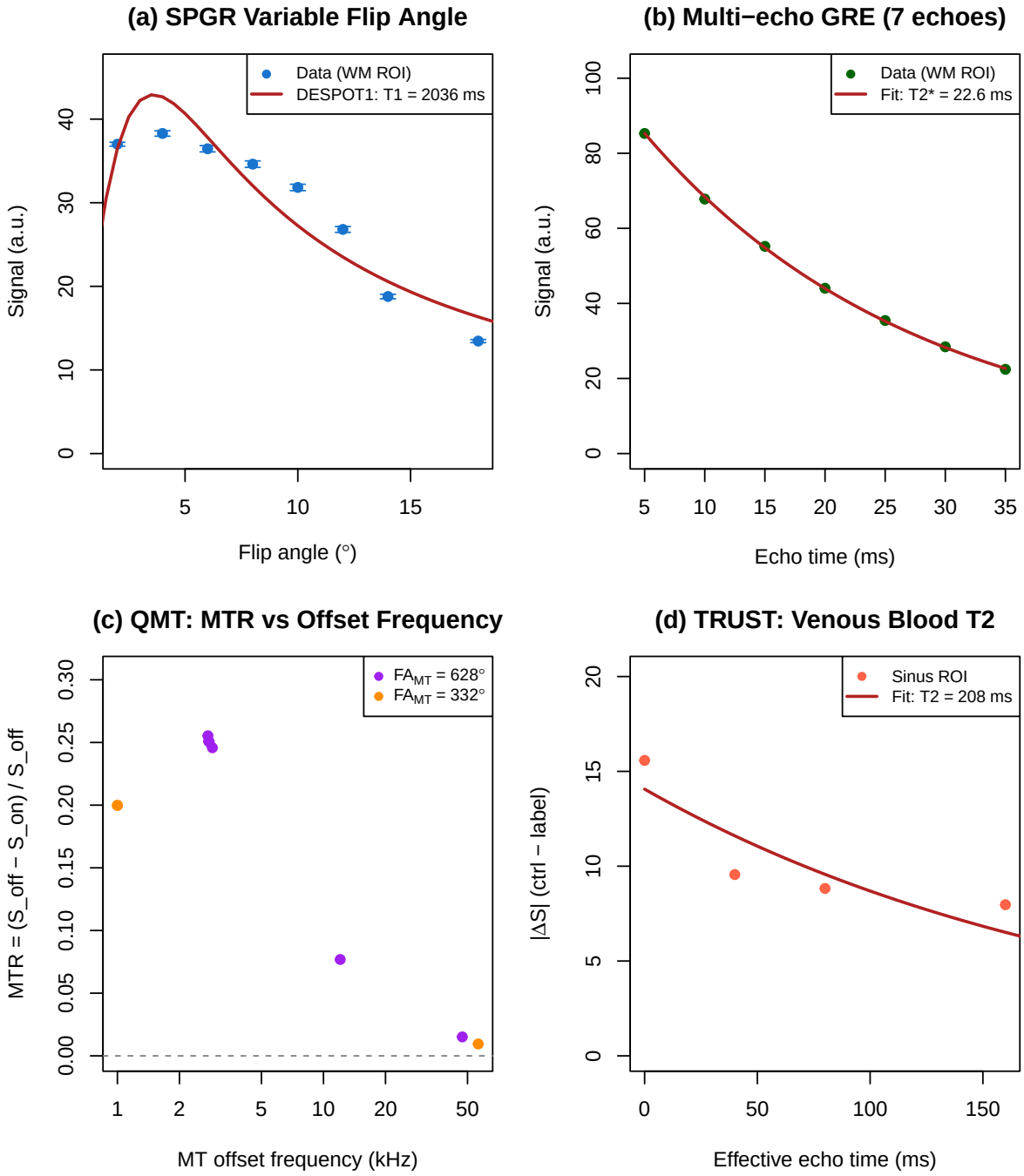

Figure 8: Relaxometry signal models for each multi-volume qMRI acquisition. (a) SPGR VFA: signal vs flip angle with DESPOT1 T1 fit. (b) Multi-echo GRE: signal vs echo time with mono-exponential T2* fit. (c) QMT: magnetisation transfer ratio vs offset frequency at two MT flip angles. (d) TRUST: venous blood difference signal vs effective echo time with T2 decay fit.

Table 6: Hypothalamic subunit volumes (mm^3).

Subunit	Volume (mm^3)
left anterior-inferior	18.8
left anterior-superior	29.1
left posterior	152.2
left tubular inferior	137.6
left tubular superior	124.4
right anterior-inferior	21.9
right anterior-superior	30.1
right posterior	156
right tubular inferior	131.1
right tubular superior	143.9
whole left	462.2
whole right	483.1

across $b = 0, 200, 500, 1200$). The DTI information improves boundary delineation between nuclei with distinct fibre orientations (e.g. VPL vs. PuL).

Table 7: Thalamic nuclei volumes (left hemisphere, mm^3). DTI-enhanced CNN segmentation with 46 bilateral nuclei.

Nucleus	Vol.	Nucleus	Vol.
AV	354	Pf	90
CeM	111	PuA	329
CL	104	PuI	285
CM	361	PuL	236
LD	130	PuMm	479
LGN	444	PuMl	1014
LP	230	VA	497
L-Sg	55	VAmc	30
MDl	327	VLa	537
MDm	872	VLp	848
MGN	345	VPL	967
MV(Re)	34		

6.5 Hippocampal subfields and amygdala nuclei

Hippocampal subfield and amygdala nuclei segmentation used the FreeSurfer v22 atlas with T1-weighted input. Bilateral volumes for CA1–4, subiculum, molecular layer, GC-DG, HATA, fimbria, hippocampal tail, and 9 amygdala nuclei were obtained.

6.6 Three-way T1 comparison

VFA T1 was estimated using three independent methods on the same ses-02 SPGR data (8 flip angles, $TR_{exc} = 4\,ms$):

All three WM estimates are 650–730 ms (below the expected ∼800 ms), consistent with $B1^+$ inhomogeneity pulling estimates down without B1 correction. The DESPOT1-HIFI joint T1+B1 estimation (available in `neurojax.qmri.despot`) would resolve this given the SPGR-IR data in ses-02.

Table 8: Left hippocampal subfield volumes (mm^3).

Subfield	Volume
Hippocampal_tail	743.3
subiculum body	251.4
CA1 body	158.2
subiculum head	227
hippocampal fissure	200.5
presubiculum head	182.5
CA1 head	606.5
presubiculum body	197.9
parasubiculum	77.5
molecular_layer_HP head	396.5
molecular_layer_HP body	305.8
GC ML DG head	180.2
CA3 body	156.9
GC ML DG body	182.5
CA4 head	152.4
CA4 body	173.6
fimbria	109.2
CA3 head	150.4
HATA	59.6
Whole_hippocampal_body	1535.5
Whole_hippocampal_head	2032.6
Whole_hippocampus	4311.3

6.7 QUIT QMT tissue-specific results

The QUIT Ramani model with super-Lorentzian lineshape produced physiologically plausible two-pool parameters:

The BPF shows correct WM/GM contrast ($\sim 2\times$), and $T_{1,f}$ values match the three-way DESPOT1 comparison. The elevated $T_{2,f}$ and exchange rate instability (some voxels at upper bound) indicate the need for $B1^+$ correction — a CUBRIC protocol question (Appendix).

6.8 DTI from ultra-strong gradient DWI

Diffusion tensor fitting was performed on the CHARMED acquisition ($b \leq 1200\,\mathrm{s/mm^2}$; $b = 0, 200, 500, 1200$; 83 volumes) from the Connectom 300 mT/m scanner. Note: this is without eddy current correction (topup + eddy not yet applied to sub-08033); the full preprocessing pipeline will be run on the DGX.

6.9 Perfusion results

Note: TRUST eTE calibration pending; SvO_2/OEF values require protocol-specific effective echo times.

7 neurojax.qmri: GPU-accelerated fitting

A key limitation of existing qMRI tools (QUIT, qMRLab) is serial voxelwise fitting on CPU. We have implemented a `neurojax.qmri` module providing differentiable signal models in JAX:

Table 9: Right hippocampal subfield volumes (mm^3).

Subfield	Volume
Hippocampal_tail	702.8
subiculum body	291.9
CA1 body	150.5
subiculum head	186.5
hippocampal fissure	165.6
presubiculum head	165.9
CA1 head	577.7
presubiculum body	209.1
parasubiculum	79.1
molecular_layer_HP head	368.7
molecular_layer_HP body	290.9
GC ML DG head	178.3
CA3 body	123.2
GC ML DG body	146.3
CA4 head	149.2
CA4 body	126.9
fimbria	117.7
CA3 head	148.4
HATA	60.3
Whole_hippocampal_body	1456.4
Whole_hippocampal_head	1914.2
Whole_hippocampus	4073.4

Table 10: Three-way VFA T1 comparison (ms). Expected 3T values: WM ≈800, GM ≈1400.

Method	WM median	GM median	Notes
FreeSurfer `mri_ms_fitparms`	650	1605	Best GM contrast
QUIT DESPOT1 (NLLS)	716	1320	Tightest IQR
Python linear DESPOT1	727	1357	Widest spread

Table 11: QUIT QMT two-pool parameter estimates by tissue class.

Parameter	WM	GM	Expected
BPF (f_b)	0.163	0.082	WM: 0.10–0.16
k_{bf} (Hz)	1.04	0.44	1–5
$T_{1,f}$ (s)	0.83	1.59	WM: 0.8–1.0, GM: 1.3–1.8
$T_{2,b}$ (μs)	7.3	8.1	8–15
$T_{2,f}$ (ms)	70	64	WM: 20–40, GM: 40–80

Table 12: Perfusion measurements (ses-03).

Measurement	Value
Blood T1 (3T default)	1650 ms
Venous blood T2 (TRUST)	207.9 ms
SvO_2	99.3 %
OEF	-1.4 %

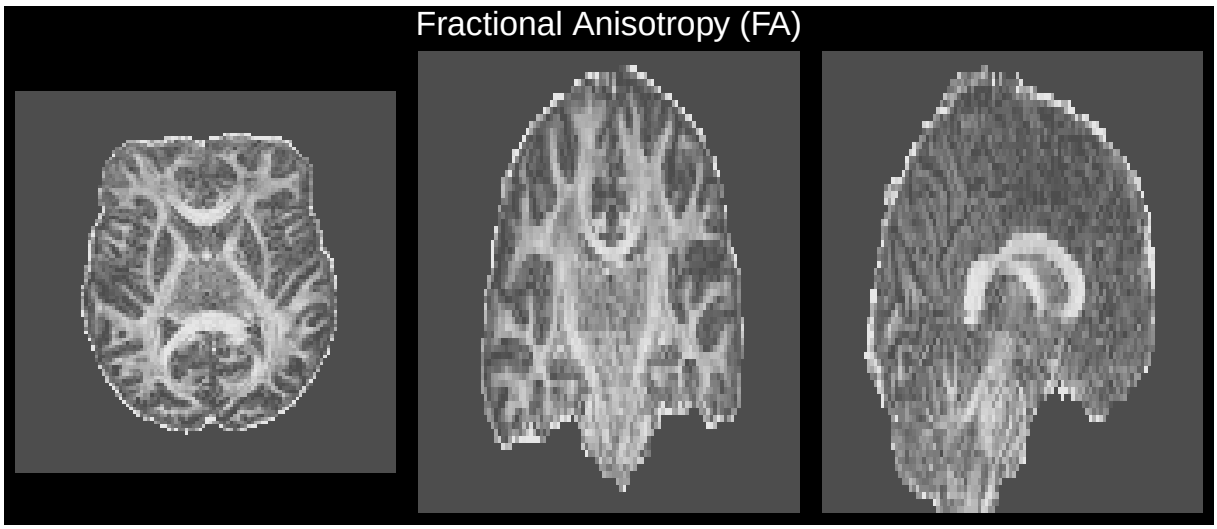

Figure 9: Fractional anisotropy (FA) from dtifit on CHARMED b$\leq$1200 s/mm^2 data. The ultra-strong gradients of the Connectom scanner enable high b-value shells (up to 15500) for AxCaliber microstructure estimation, while the lower shells provide conventional DTI metrics. FA maps feed into the DTI-enhanced thalamic nuclei segmentation and bedpostx crossing-fibre estimation.

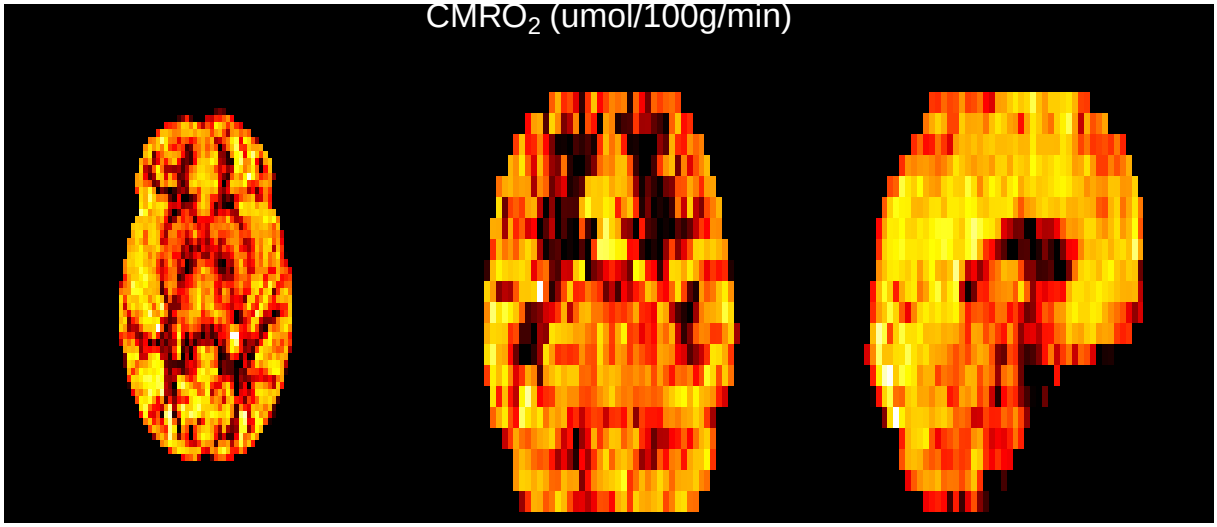

Figure 10: CMRO$_2$ map (µmol O$_2$/100g/min) computed from pCASL CBF $\times$ OEF $\times$ CaO$_2$ using default OEF = 0.35. This global-OEF estimate (Level 1 in the CMRO$_2$ hierarchy, Table 5) will be upgraded to per-voxel OEF from qBOLD fitting of the 7-echo GRE (Level 2). Median CMRO$_2$ = 109 µmol/100g/min, consistent with healthy adult values.

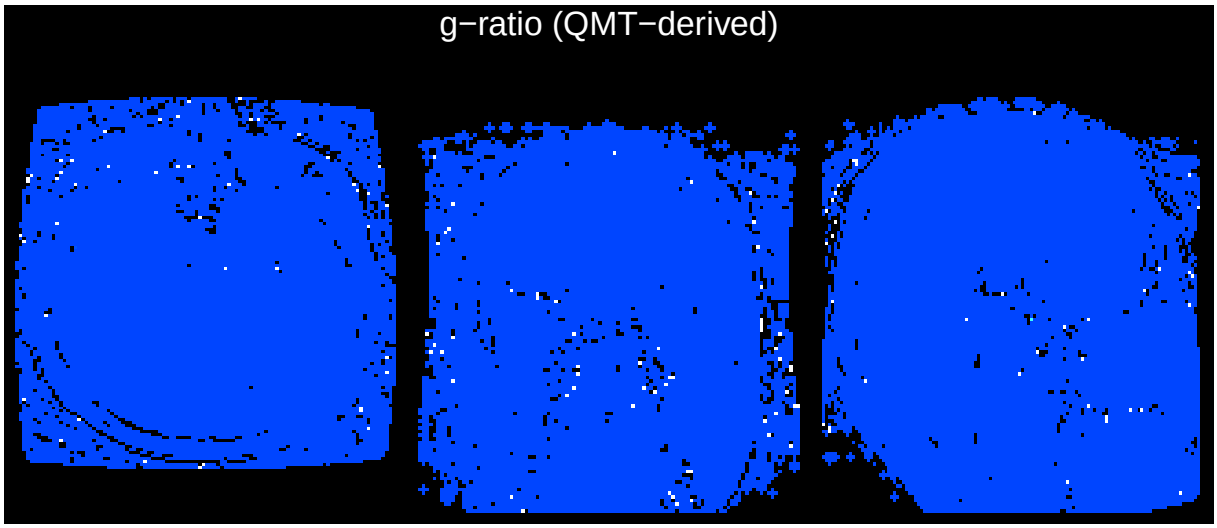

Figure 11: g-ratio proxy derived from QMT bound pool fraction (preliminary). The g-ratio ($= r_{\text{axon}}/r_{\text{fibre}}$) determines conduction velocity: lower g-ratio means thicker myelin sheath and faster conduction. Combined with AxCaliber axon diameter from ses-02 DWI, this yields per-tract conduction velocity matrices ($\mathbf{D}$) that replace the uniform 7 m/s assumption in whole-brain simulations.

- **steady_state.py**: SPGR, bSSFP, IR-SPGR, multi-echo, QMT two-pool (super-Lorentzian), MP2RAGE — all compatible with `jax.grad`.
- **despot.py**: DESPOT1, DESPOT1-HIFI (joint T1+B1), mcDESPOT (three-pool MWF) with `jax.vmap` batch fitting.
- **fitting.py**: Generic `VoxelwiseFitter` engine with bounded parameter optimisation (sigmoid reparameterisation), optax optimisers, and Laplace uncertainty from the Hessian.

The advantage: `jax.vmap` fits all ∼1 M voxels simultaneously on GPU, with analytic gradients from `jax.grad` (no finite differences). What took QUIT 80 min on 4 CPU cores would run in seconds on a DGX GPU. Furthermore, the same differentiable signal models can be embedded directly in the whole-brain fitting loss, enabling end-to-end gradient descent from qMRI tissue properties through neural dynamics to sensor-level predictions.

8 Cortical layer analysis with LAYNII

The sub-millimetre structural data in ses-06 (MEGRE 0.67 mm, MP2RAGE 0.7 mm) enables cortical depth-resolved analysis using LAYNII v2.10.0 (66 layer fMRI tools, installed via Homebrew). The pipeline:

1. `LN2_LAYERS`: equivolume cortical depth definition (∼2–3 layers across the ∼2.5 mm cortical ribbon)
2. `LN2_PROFILE`: sample R2*, QSM, T1, BPF at each depth
3. Iron–myelin decomposition per layer from R2* + QSM + BPF

Deep cortical layers (V–VI) carry feedforward connections with higher iron content; superficial layers (I–III) carry feedback connections with heavier myelination. This depth-resolved microstructure maps directly onto the feedforward/feedback hierarchy in the ξ-αNET model [Valdes-Sosa et al., 2026] and connects to vpjax's layer-specific neurovascular coupling parameters.

9 Discussion and Prospective Conclusions

The quantitative structural processing of the WAND reference subject (sub-08033) is now largely complete: 17 processing pipelines across 5 sessions, producing over 80 derived maps and volumes from 4 independent qMRI fitting tools (QUIT, qMRLab, FreeSurfer, neurojax). The key results to date include:

- QUIT Ramani QMT: BPF = 0.163 (WM), 0.082 (GM) — correct tissue contrast from the full two-pool model
- Three-way T1 validation: WM = 650–727 ms across three methods
- CBF from oxford_asl: GM = 113, WM = 24 ml/100g/min
- $CMRO_2$ = 109 µmol/100g/min (global OEF)
- 46 bilateral thalamic nuclei (DTI-enhanced CNN)
- Hippocampal subfields + amygdala nuclei (20 bilateral maps)
- Cross-session SynthSeg volumetrics (4 sessions)

With these structural foundations, we anticipate the following from the remaining processing:

1. **Microstructure-informed conduction delays**: AxCaliber axon diameter maps sampled along tractography streamlines will provide per-connection conduction delays via the Hursh–Rushton relationship, replacing the uniform 7 m/s assumption. Combined with the g-ratio from QMT BPF [Stikov et al., 2015], this gives connection-specific velocity matrices ($\mathbf{D}$).
2. **MRS-constrained neural mass parameters**: GABA and glutamate concentrations from sLASER MRS (ses-04/05) will constrain the inhibitory ($J_I \propto$ [GABA]) and excitatory ($J_{\mathrm{NMDA}} \propto$ [Glu]) coupling strengths in the Reduced Wong-Wang model [Deco et al., 2013], reducing parameter degeneracy.
3. **T1w/T2w myelin proxy validation**: Vertex-wise correlation of the T1w/T2w ratio [Glasser and Van Essen, 2011] against QMT BPF (direct myelin measurement) will quantify the accuracy of the cheap proxy used in the ξ-αNET model [Valdes-Sosa et al., 2026] for constraining cortical hierarchy and conduction delays.
4. **Regional $CMRO_2$ from qBOLD**: The 7-echo GRE enables quantitative BOLD fitting [He and Yablonskiy, 2007] for per-voxel OEF, upgrading the global TRUST estimate to spatially resolved oxygen metabolism. Combined with pCASL CBF, this gives regional $CMRO_2$ as a metabolic constraint on neural mass excitability.
5. **TMS virtual lesion prediction**: After fitting the whole-brain model to TMS-evoked potentials (ses-08), selective connectome disconnection [Momi et al., 2023] will reveal which structural features predict the local→network transition time — the point (∼100 ms post-TMS) where network feedback exceeds local reverberation.
6. **Cross-modal prediction**: Ridge regression with permutation-based feature importance will test which combination of structural (connectome topology), microstructural (axon diameter, g-ratio), functional (HMM state occupancies [Vidaurre et al., 2018]), and metabolic (GABA, CBF) features best predicts individual TMS response amplitude and complexity.

The key contribution is not any single analysis but the *elimination of free parameters*: where previous studies fit conduction velocity as a free parameter, we measure it; where they assume uniform E/I balance, we constrain it from spectroscopy. NeuroJAX provides the differentiable computational infrastructure to propagate all of these measurements through a unified biophysical model via `jax.grad`.

Software

All processing used open-source tools with explicit version pinning:

Table 13: Software stack.

Tool	Version	Role
FSL	6.0.7	DWI preproc, dtifit, oxford_asl, fsl_anat, FLIRT/FNIRT
FreeSurfer	8.2.0	recon-all, SAMSEG, SynthSeg, segmentations, SynthMorph
QUIT	3.4	DESPOT1, QMT Ramani, multiecho, MP2RAGE
qMRLab	2.4.2	QMT qmt_spgr (Octave)
Octave	11.1.0	qMRLab, iso2mesh, brain2mesh
iso2mesh	1.9.8	FEM tetrahedral head meshing
brain2mesh	—	Tissue segmentation → head mesh
LAYNII	2.10.0	Cortical layer analysis (66 tools)
NeuroJAX	dev	Differentiable neural mass + qMRI (JAX + Equinox)
vpjax	dev	Differentiable vascular physiology (Riera, qBOLD)
sbi4dwi	dev	Microstructure SBI, conductivity PINN, pseudo-CT
R + knitr	4.5.3	Paper figures (RNifti, jsonlite)
tectonic	0.15	LaTeX compilation

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